PrediCT Heart Segmentation Model – Development Report

This document outlines the iterative development of the PrediCT Heart Segmentation Model for Google Summer of Code (GSoC). It details the technical modifications implemented to improve performance from a baseline Dice score of 0.67 to a final submission-ready Dice score of 0.8513 , while achieving highly optimized inference times.

Project 1: Optimized Cardiac Segmentation & Phenotyping

Developer: Shriyam Baloni

Target: Dice > 0.85 | Inference Time < 10s

Phase 1: Architectural Pivot (Attention-Gated 3D U-Net)

Modification:
Transitioned from a standard 3D U-Net to an Attention-Gated 3D U-Net architecture using the MONAI framework.

Rationale:
Standard U-Net architectures often struggle with spatial noise, assigning similar importance to non-cardiac structures such as lung tissue and bone. Attention gates enable the model to selectively suppress irrelevant regions within the thoracic cavity while emphasizing the spatial features of the heart.

Implementation:

- Integrated monai.networks.nets.AttentionUnet with four levels of depth
- Utilized 3D spatial dimensions with channel progression (16, 32, 64, 128) to maintain a lightweight memory footprint on Apple M2 hardware

Phase 2: Addressing Class Imbalance (Custom Focal Tversky Loss)

Modification:
Replaced the standard Dice Loss with a manually implemented Focal Tversky Loss (FTL).

Rationale:
In cardiac CT imaging, background voxels significantly outnumber target voxels. Standard loss functions tend to produce conservative models that fail to capture ambiguous heart boundaries. FTL incorporates a focal parameter (γ) that down-weights easy examples and prioritizes difficult boundary regions.

Implementation:

- Defined a custom FocalTverskyLoss class inheriting from MONAI's TverskyLoss
- Set parameters $\alpha = 0.7$ (to penalize false positives such as rib leakage) and $\gamma = 1.3$ (to enhance boundary refinement)

Phase 3: Regional Focus (ROI & Mediastinal Cropping)

Modification:
Implemented center spatial cropping during preprocessing to isolate the anatomical region of interest.

Rationale:
Earlier iterations exhibited mediastinal leakage, where thoracic musculature was incorrectly segmented as cardiac tissue. Restricting the field of view (FOV) to the central thoracic region significantly reduced false-positive regions prior to inference.

Implementation:

- Applied CenterSpatialCropd with a 256×256 region of interest
- Balanced sufficient anatomical context (lungs and spine) while maintaining Dice performance above the target threshold

Phase 4: Topological Refinement (Erosion–LCC–Dilation)

Modification:
Introduced a three-stage morphological post-processing pipeline.

Rationale:
Even at high Dice scores, small topological connections may incorrectly link the heart mask to surrounding tissues. This refinement ensures a single, anatomically consistent segmentation output.

Implementation:

1. **Binary Erosion:** Reduced the mask by three iterations to eliminate thin connections to adjacent structures
2. **Largest Connected Component (LCC):** Retained only the largest 3D component corresponding to the heart
3. **Binary Dilation:** Restored the mask to its original extent by applying three dilation iterations

Phase 5: Results and Performance Analysis

The final model achieves a strong balance between segmentation accuracy and computational efficiency. This pipeline demonstrates a practical trade-off between accuracy and computational efficiency, making it suitable for scalable radiomics and calcium scoring workflows in large clinical datasets.

Comparison Table

Metric	TotalSegmentator (Baseline)	PrediCT Model (Ours)
Dice Score	1.0 (Reference)	**0.8513**
Inference Time	~180.00 s	**9.45 s**